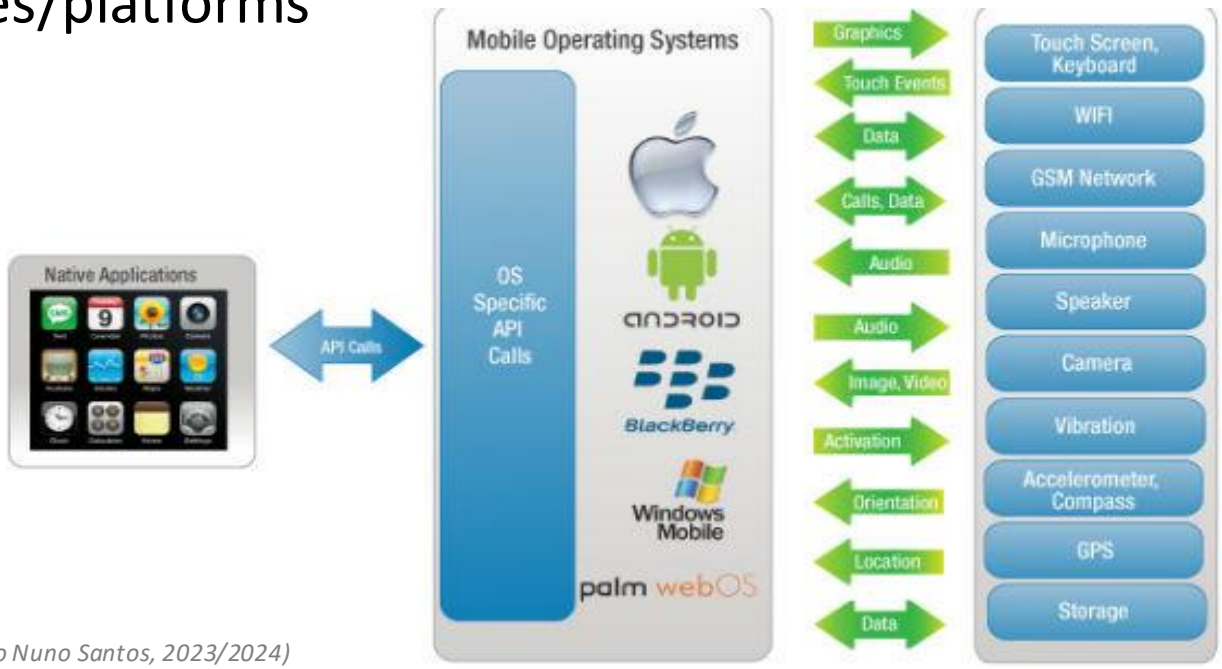
Mobile Architectures

Mobile Systems

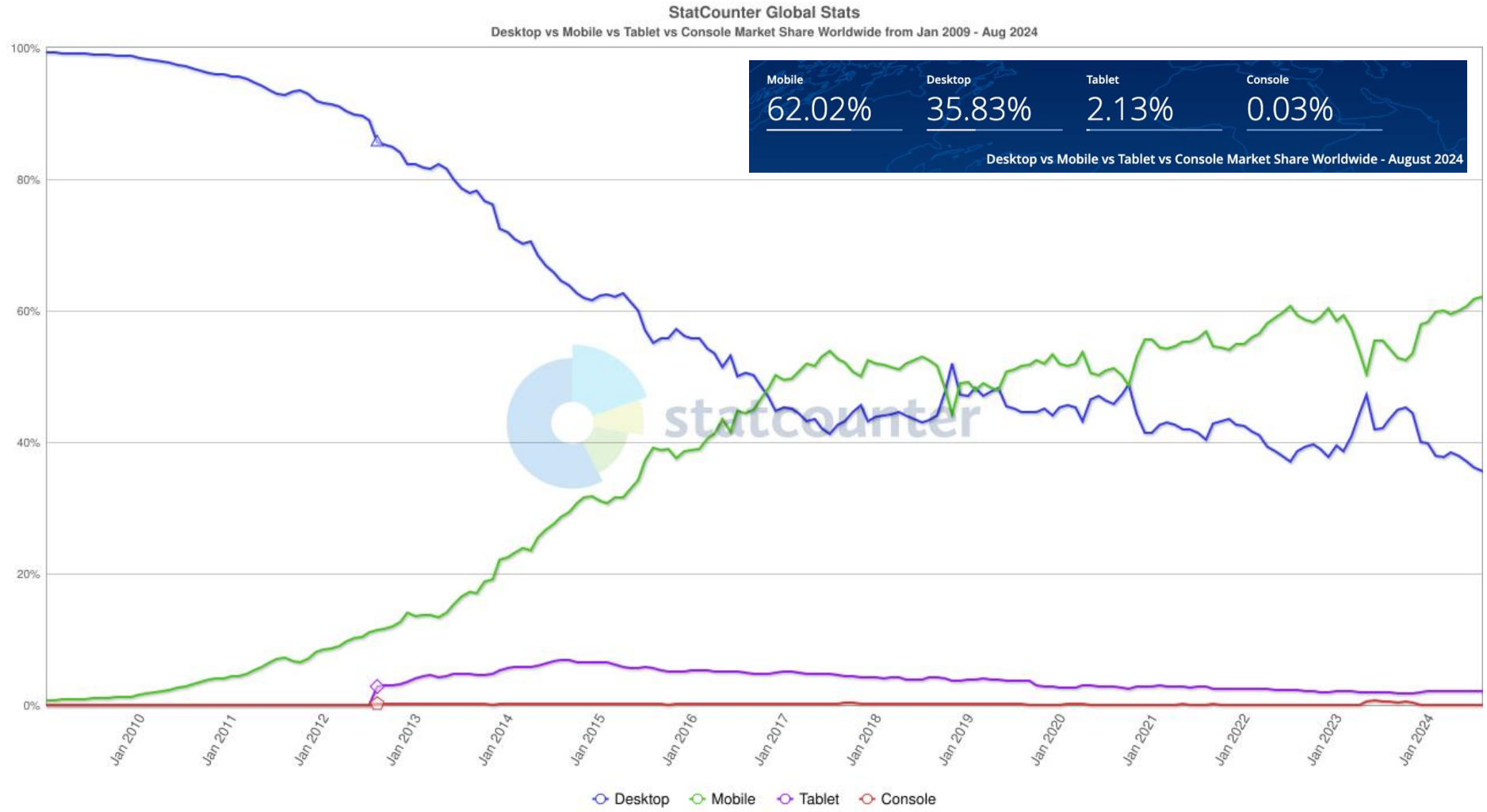
Mobile operating systems

- Operating system
  - Software layer that virtualizes the hardware, facilitating and maximizing its use

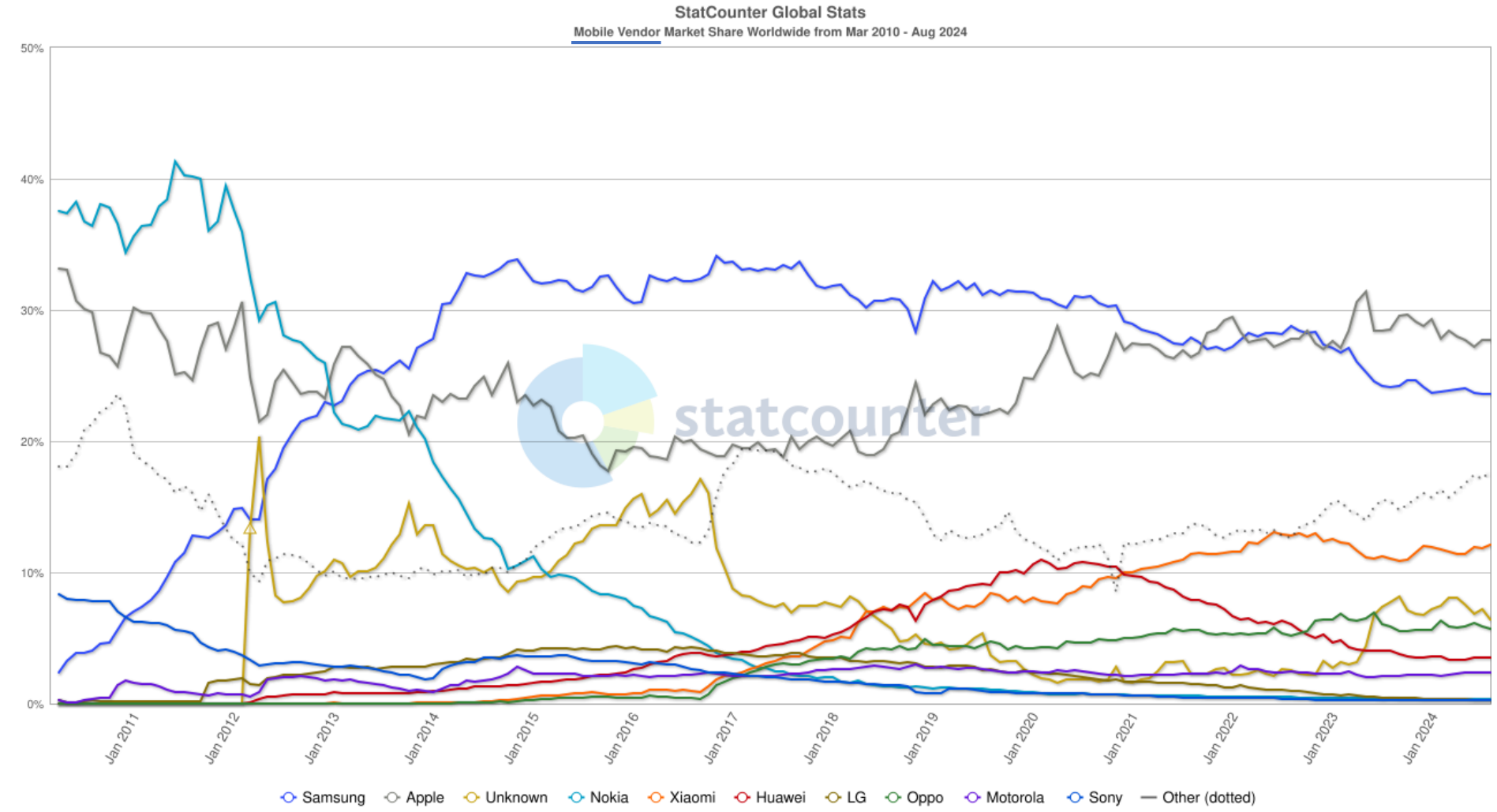
- Mobile operating system
  - It is an operating system suited to the specificities of mobile devices/platforms



Market share



Manufacturers



Mobile operating systems

- Symbian OS
- RIM BlackBerry OS
- iOS (Apple)
- Windows CE
- Windows Mobile
- Windows Phone



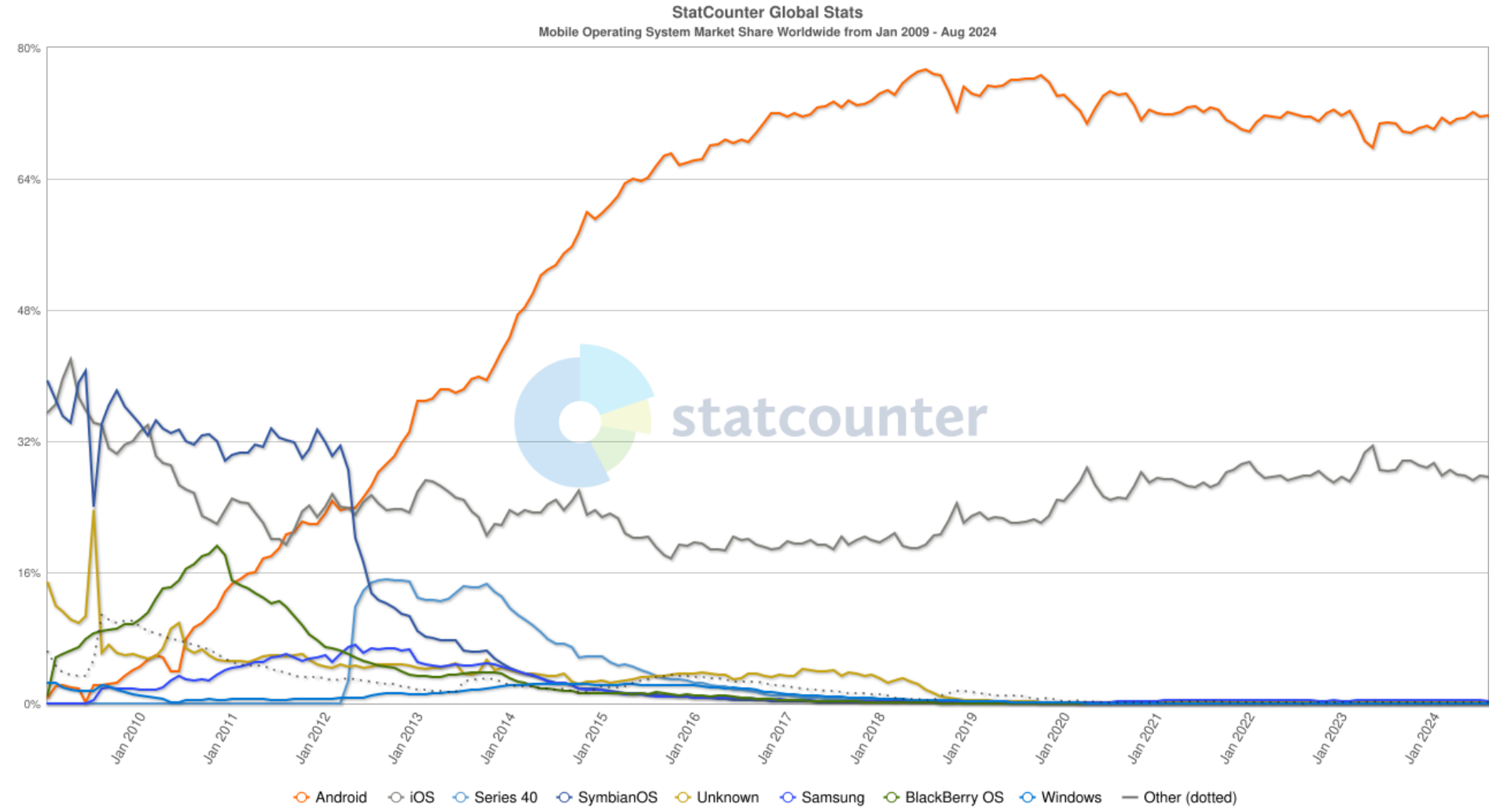
- Android
- Firefox OS
- KaiOS
- LiMo
- Maemo
- MeeGo
- Bada
- Mer
- Tizen
- Open Web OS/ webOS
- Jolla Sailfish OS/ Sailfish UI
- Ubuntu Touch
- …
- HarmonyOS



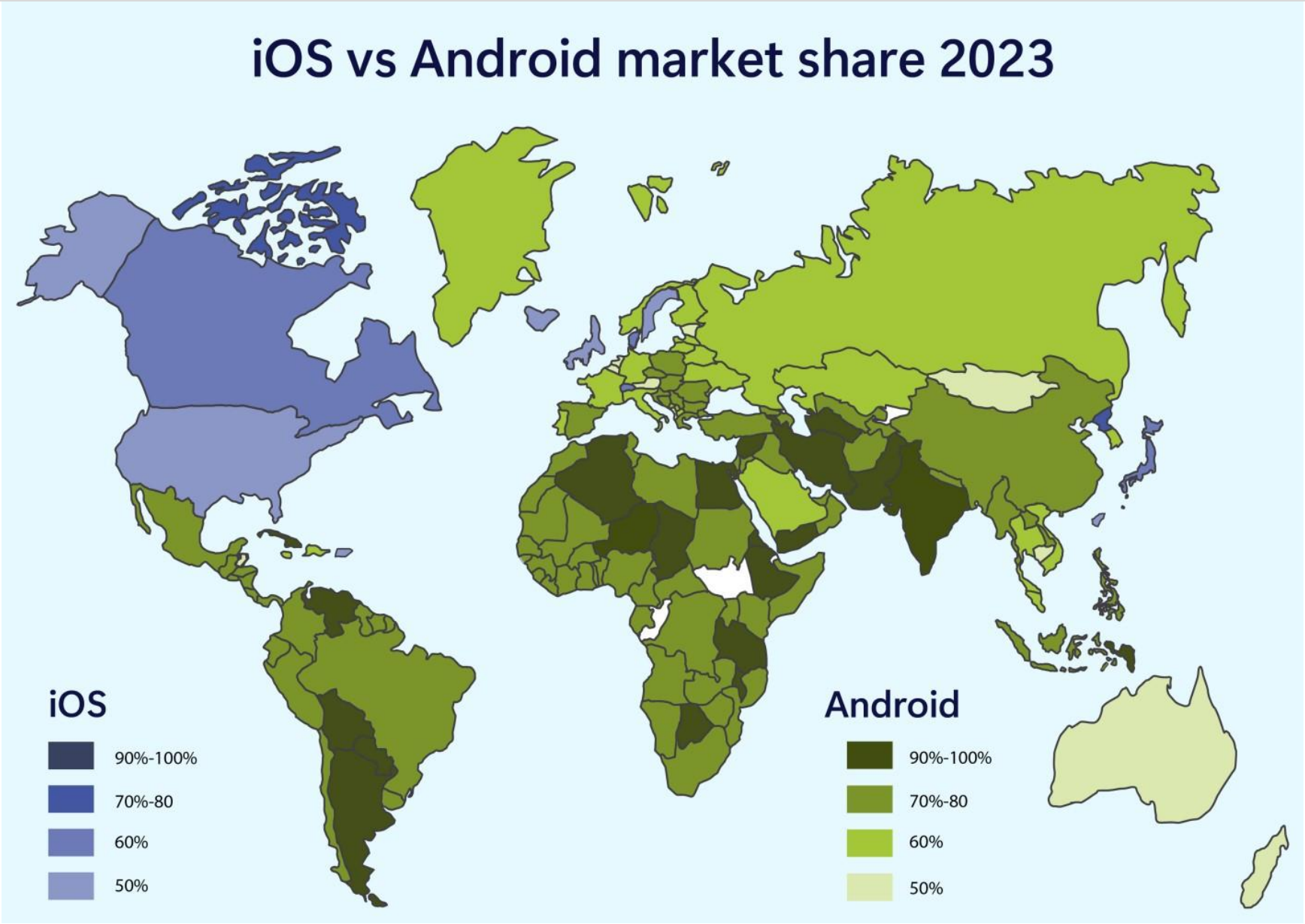
Mobile OS

Feature	Android	iOS	HarmonyOS	Tizen	KaiOS	Sailfish OS Aurora OS	Ubuntu Touch	Mobian[1]	Plasma Mobile	PureOS	PostmarketOS
Developed by	Google, Open Handset Alliance	Apple Inc.	Huawei	Linux Foundation, Tizen Association, Samsung, Intel	KaiOS Technologies Inc.(TCL)	Sailfish Alliance, Mer, Jolla and Sailfish community contributors	UBports and Ubuntu community contributors (previously Canonical Ltd.)	Debian on Mobile Team	KDE and Blue Systems	Purism	PostmarketOS community
Market share[2][3][4]	71.90%	27.33%	4%	0.22%	0.14%	N/A	N/A	N/A	N/A	N/A	N/A
License	Base system is open source, but many devices use proprietary drivers for hardware support, and most Android operating systems include Proprietary apps (such as Google Play and other Google apps). [5] [6]	Proprietary, open source kernel and core	Proprietary except for open-source components	Partial; both proprietary and open-source components, assorted licenses	Proprietary except for open source kernel patches (formerly the MPL 2.0 B2G OS)	Free and open-source, but the UI and the SDK are proprietary and closed source	Free and open-source, mainly the GPL[7]	Free and open-source, mainly the GPL	Free and open-source, mainly the GPL	Free and open-source, mainly the GPL	Free and open-source, GPL
Current version	14	17.5.1	2.0.0.66 (Canary2)	4.0.0.7	3.1	4.5.0.24	20.04 OTA-4	20210516	5.24.3	10.0	21.12 Service Pack 3
Development version	15	18 Beta 3/PB1	HarmonyOS NEXT DP2 5.0.0	8.0 M2	3.2	Unknown	Unknown	20220612	Unknown	Unknown	N/A
Current version release dates	October 4, 2023; 11 months ago	May 20, 2024; 3 months ago	September 26, 2023; 2 months ago	May 30, 2020; 4 years ago[8]		September 12, 2023; 11 months ago[9]	February 18, 2022; 2 years ago	May 16, 2021; 3 years ago	March 8, 2022; 2 years ago	October 2, 2021; 2 years ago	March 13, 2022; 2 years ago

Worldwide


StatCounter Global Stats
Mobile Operating System Market Share Worldwide from Jan 2009 - Aug 2024
80%
64%
48%
32%
16%
0%
statcounter
Jan 2010
Jan 2011
Jan 2012
Jan 2013
Jan 2014
Jan 2015
Jan 2016
Jan 2017
Jan 2018
Jan 2019
Jan 2020
Jan 2021
Jan 2022
Jan 2023
Jan 2024
Android
iOS
Series 40
SymbianOS
Unknown
Samsung
BlackBerry OS
Windows
Other (dotted)

Dominant OS



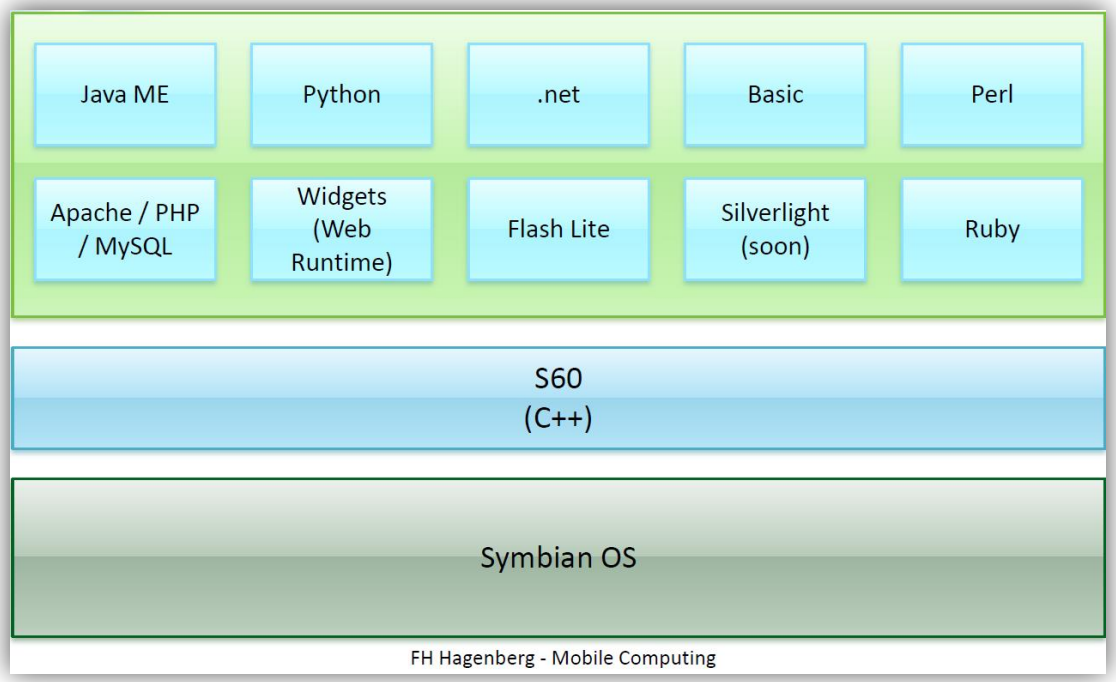
https://www.linkedin.com/pulse/ios-vs-android-market-share-zamaqo-kjizf/

Symbian OS

- Evolution of PSION's EPOC system (1989-2000) designed for PDA
- *Symbian* was created in 1998
  - Joint venture between *Psion* and some mobile phone manufacturers (*Ericsson*, *Motorola* and *Nokia*)
- The first version of Symbian OS came out in 2001
  - Nokia 9210
- In 2008 Nokia acquired *Symbian Software Limited*
  - *Symbian Foundation* was created – autonomous and non-profit entity to manage Symbian OS
  - In November 2010, the project was abandoned, leaving only support and development for specific markets
- Preemptive, single-user, multitasking operating system with memory protection
- http://www.symbian.org

Symbian OS

- Development…
  - Programming was essentially carried out in C++
    - Symbian C++
  - Subsystems were developed to support other languages/development platforms



RIM BlackBerry

- RIM – *Research In motion*
  - Canada
  - 1996
- Main objective: “personal digital assistant”
  - Email
  - Calendar
  - Tasks
  - ***Security***
- Development
  - Essentially JAVA
  - *Web Service extensions* (*Server* + *Client*)
- Evolution (more recent years)
  - Android system was adopted for their devices
  - Dedicated to business and security services

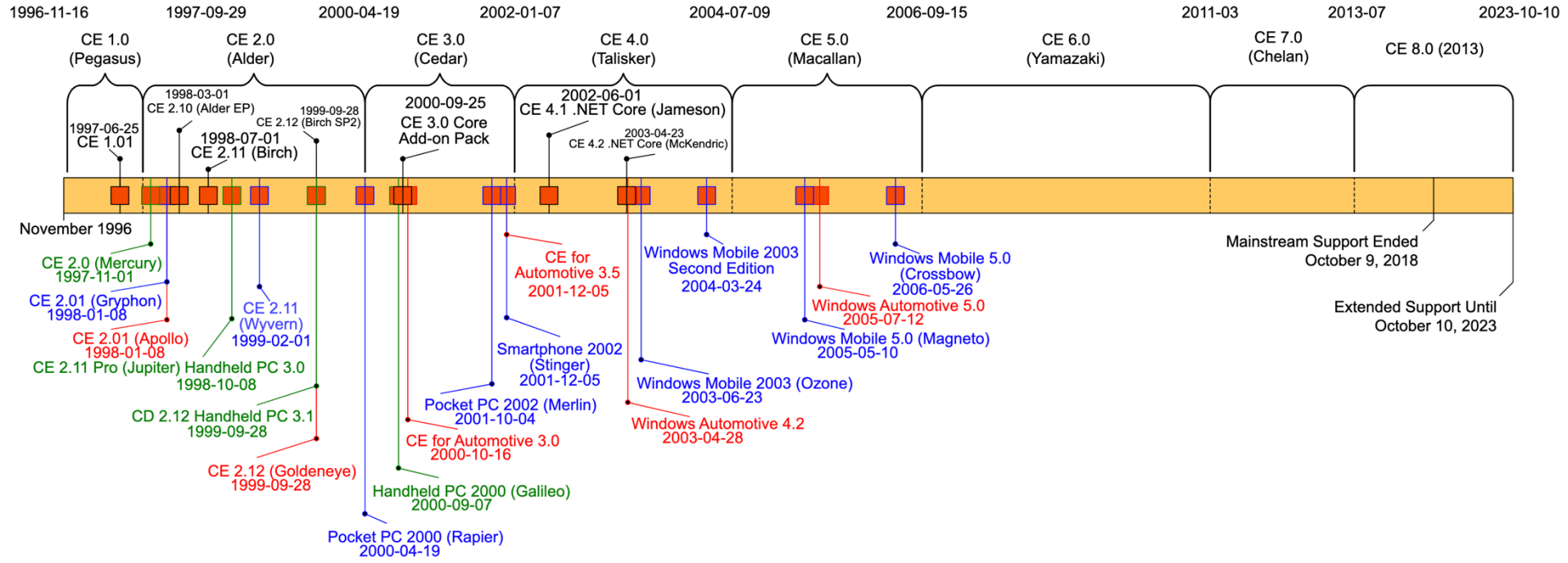


Windows CE



- Windows Embedded Compact
  - Some call it “Windows Compact Edition”
  - Microsoft says “Compact, Connectable, Compatible, Companion, and Efficient ”
- Developed by Microsoft for embedded systems and/or limited resources
  - Goal: Provide “similar” features to those found on a Windows computer
- 1st version: 1996
- Last version: 8
  - Windows Embedded Compact 2013

Windows CE


1996-11-16
1997-09-29
2000-04-19
2002-01-07
2004-07-09
2006-09-15
2011-03
2013-07
2023-10-10
CE 1.0 (Pegasus)
CE 2.0 (Alder)
CE 3.0 (Cedar)
CE 4.0 (Talisker)
CE 5.0 (Macallan)
CE 6.0 (Yamazaki)
CE 7.0 (Chelan)
CE 8.0 (2013)
1997-06-25 CE 1.01
1998-03-01 CE 2.10 (Alder EP)
1998-07-01 CE 2.11 (Birch)
1999-09-28 CE 2.12 (Birch SP2)
2000-09-25 CE 3.0 Core Add-on Pack
2002-06-01 CE 4.1 .NET Core (Jameson)
2003-04-23 CE 4.2 .NET Core (McKendric)
November 1996
CE 2.0 (Mercury) 1997-11-01
CE 2.01 (Gryphon) 1998-01-08
CE 2.01 (Apollo) 1998-01-08
CE 2.11 Pro (Jupiter) Handheld PC 3.0 1998-10-08
CE 2.11 (Wyvern) 1999-02-01
CD 2.12 Handheld PC 3.1 1999-09-28
CE 2.12 (Goldeneye) 1999-09-28
Pocket PC 2000 (Rapier) 2000-04-19
Handheld PC 2000 (Galileo) 2000-09-07
CE for Automotive 3.0 2000-10-16
Pocket PC 2002 (Merlin) 2001-10-04
Smartphone 2002 (Stinger) 2001-12-05
CE for Automotive 3.5 2001-12-05
Windows Automotive 4.2 2003-04-28
Windows Mobile 2003 (Ozone) 2003-06-23
Windows Mobile 2003 Second Edition 2004-03-24
Windows Mobile 5.0 (Magneto) 2005-05-10
Windows Automotive 5.0 2005-07-12
Windows Mobile 5.0 (Crossbow) 2006-05-26
Mainstream Support Ended October 9, 2018
Extended Support Until October 10, 2023

Windows Mobile

- Based on *Windows CE*
  - Windows *Mobile 6* is based on *Windows CE 5.2*
- Used on cell phones/*smartphones*
- *Standard* and *Pro* versions
  - The *Pro version* has touchscreen support
- Latest version: 6.5.5

- Development
  - Various languages
    - Initially, the most common was C/C++ (Win32-API, MFC)
    - After, development began to be supported through the .NET Compact Framework
      - Subset of the .NET Framework
        - Version of the .NET Framework adapted to devices with limited resources
        - Continues to offer services not normally associated with this type of equipment
          - For example, SQL – using the Microsoft SQL Server CE version
      - C#
      - VB.NET

Windows Phone

- *Windows Phone 7*
  - Based on *Windows CE 6.0*
- *Windows Phone 8* and *8.1*
  - Based on *Windows NT + Windows 8*
- *Windows 10 Mobile*

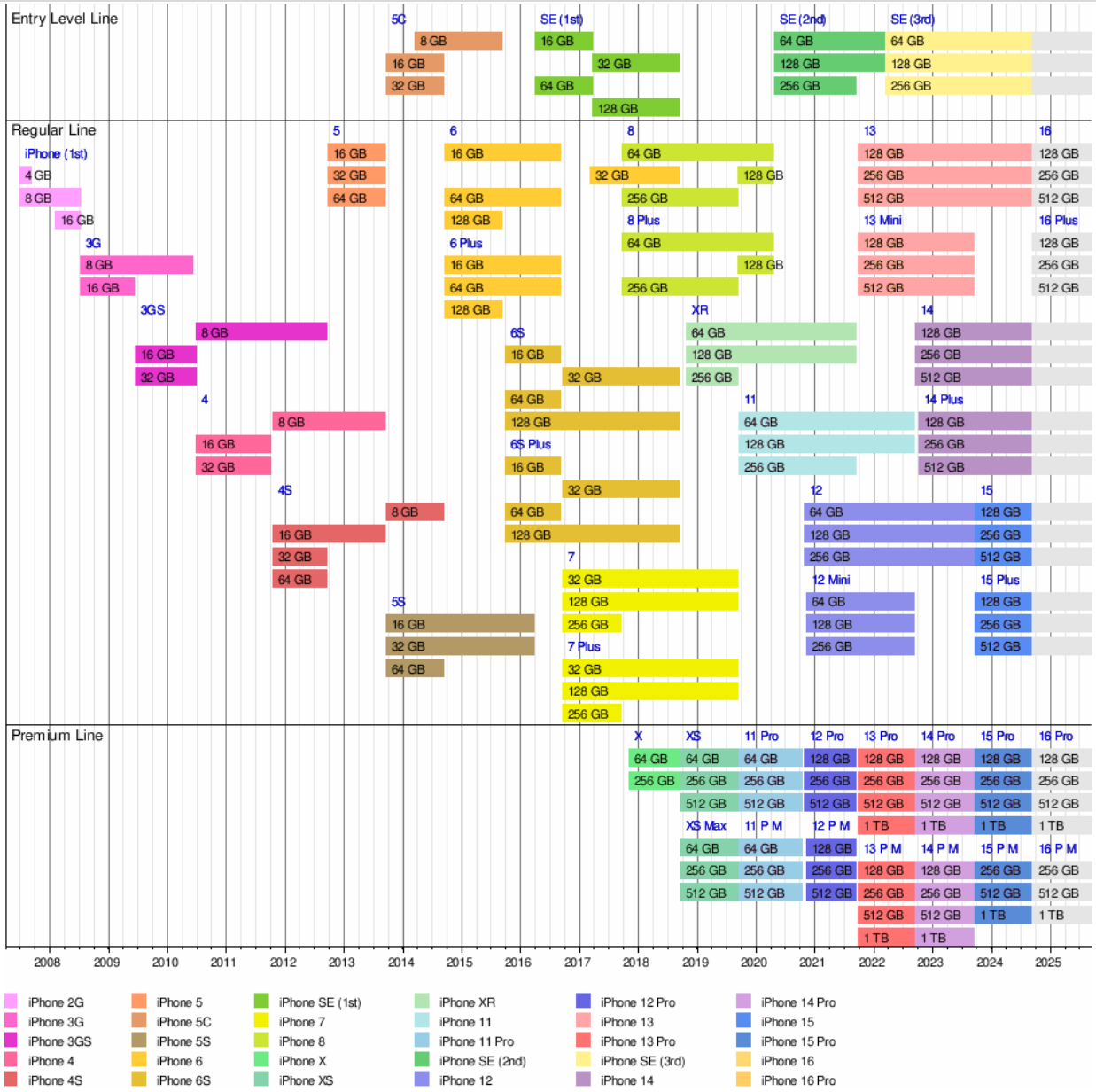
- Pleasant interface but failed to gain market share
- Main advantage: integration with business applications and Office
- Cortana assistant was introduced to respond to the competition (Siri, Google Now)

- Development
  - Based on two technologies
    - Silverlight
    - XNA
  - Development platform: Visual Studio
    - There was a version of Visual Studio Express prepared for developing applications for Windows Phone
  - Base language: C#
  - Applications were made available through the Windows Phone Marketplace/App hub
    - 30% to Microsoft
    - $99/year

iOS

- Designed for *iPhone* and *iPod Touch* from Apple
  - Strong relationship with *hardware*
  - Simplicity of use
  - Stability
- iOS devices are considered the main reference in *smartphones* and *tablets*
- Initially based on OS X 10.5 (*Leopard*)
- Multitasking support began with the release of version 4
  - In previous versions it was already possible to multitask between some specific applications (e. g., music)
- Latest version released: iOS 17.6.1
  - *The iOS 18 and the new iPhones will be available on September 20, 2024*
    - *https://www.apple.com/apple-events/*
  - In recent versions, there has been an approximation in the provision of functionalities until now associated with other systems (mainly Android)
  - iPadOS appeared with the launch of iOS 13

iPhone



https://en.wikipedia.org/wiki/Template:Timeline_of_iPhone_models

iOS (iPhone, iPad)

- Development
  - Applications distributed through the 'App Store'
    - Free applications
    - Paid applications
      - 30% to Apple
    - $99/year to subscribe to iOS Developer Program
  - Necessary tools and requirements:
    - SDK
    - Xcode – development environment
      - Includes iPhone Simulator
    - Languages: C, C++, Objective-C, **Swift**, **SwiftUI**
    - Computer with macOS
    - iPhone, iPad, Apple Watch, Apple TV

Android



Android is a trademark of Google LLC.

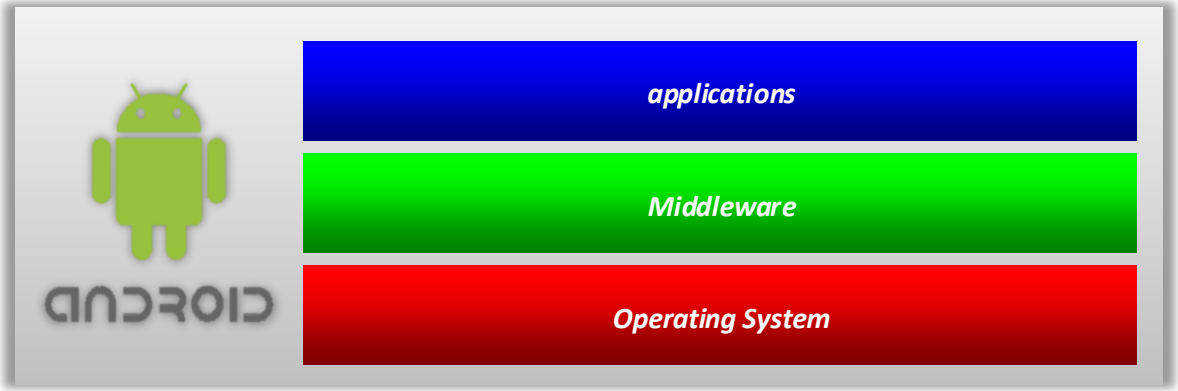
- Open-source initiative developed by a group of companies, headed by Google, which was named Open Handset Alliance

  "Today, there are 1.5 billion television sets in use around the world. 1 billion people are on the Internet. But nearly 3 billion people have a mobile phone, making it one of the world's most successful consumer products" (**November 5, 2007**)

  http://www.openhandsetalliance.com/oha_overview.html

Android

- Software suite that includes the operating system, *middleware* and applications
  - Linux-based (initially, version 2.6)
    - From version 4 of *Android*, version 3 of Linux started to be used
    - From version 7 of *Android*, version 4 of Linux started to be used
    - From version 11 of *Android*, version 5 of Linux started to be used
    - From version 14 of Android, version 6 of Linux started to be used
  - Support for the latest applications and technologies in the area of mobile devices (photography, video, GPS, compass, accelerometer, games, OpenGL, SQLite, ... and cell phone 😀)



Android

- Google has followed a policy of launching its “own” device to support the presentation of a new version of the Android system (with significant changes)
  - *Nexus One* (manufactured by HTC)
    - The way chosen for commercialization undermined the success of the project
    - Android 2.1
  - *Nexus S* (made by Samsung)
    - The commercialization model followed in the second model was more successful
    - In addition to the innovations in terms of the Android system, it served to present a set of new technologies (e. g., NFC, curved screen)
    - Android 2.3
  - (Samsung) Galaxy Nexus
    - Android 4.0
  - Nexus 7 (Tablet - ASUS)
    - Android 4.1
  - Nexus 4 (LG)
    - Android 4.2
  - Nexus 7 2013 (Tablet - ASUS)
    - Android 4.3
  - Nexus 5 (LG)
    - Android 4.4



Android

- Nexus 6 (Motorola)
  - Android 5.0
- Nexus 9 (Tablet - LG)
  - Android 5.0
- Nexus 5X (LG)
  - Android 6.0
- Nexus 6P (Huawei)
  - Android 6.0



Android – Google Pixel phones

Model	Announced	Release		Support		Lifespan[a]
		Date	With OS	Date[1]	Final OS[2]	
Pixel / XL	October 4, 2016	October 20, 2016	7.1 Nougat	December 2, 2019	10	3 years, 1 month
Pixel 2 / XL	October 4, 2017	October 17, 2017	8.0 Oreo	December 7, 2020	11	3 years, 1 month
Pixel 3 / XL	October 9, 2018	October 18, 2018	9 Pie	June 6, 2022	12	3 years, 7 months
Pixel 3a / XL	May 7, 2019	May 15, 2019		September 5, 2022	12.1	3 years, 3 months
Pixel 4 / XL	October 15, 2019	October 23, 2019	10	February 6, 2023	13	3 years, 3 months
Pixel 4a	August 3, 2020	August 20, 2020		November 6, 2023		3 years, 2 months
Pixel 4a (5G)		November 5, 2020	11	February 5, 2024	14	3 years, 3 months
Pixel 5	September 30, 2020	October 15, 2020				3 years, 3 months
Pixel 5a	August 17, 2021	August 26, 2021		August 26, 2024		3 years
Pixel 6 / Pro	October 19, 2021	October 28, 2021	12	At least October 2026	—	2 years, 10 months
Pixel 6a	May 11, 2022	July 21, 2022		At least July 2027		2 years, 1 month
Pixel 7 / Pro	October 6, 2022	October 13, 2022	13	At least October 2027		1 year, 11 months
Pixel 7a	May 10, 2023	May 10, 2023		At least May 2028		1 year, 4 months
Pixel Fold		June 20, 2023		At least June 2028		1 year, 2 months
Pixel 8 / Pro	October 4, 2023	October 12, 2023	14	At least October 2030		10 months
Pixel 8a	May 7, 2024	May 14, 2024		At least May 2031		4 months
Pixel 9	August 13, 2024	August 22, 2024		At least September 2031		0 months
Pixel 9 Pro / Pro XL	August 13, 2024	August 22, 2024				
Pixel 9 Pro Fold	August 13, 2024	September 4, 2024				

https://en.wikipedia.org/wiki/Google_Pixel

Versions





2022.08.15

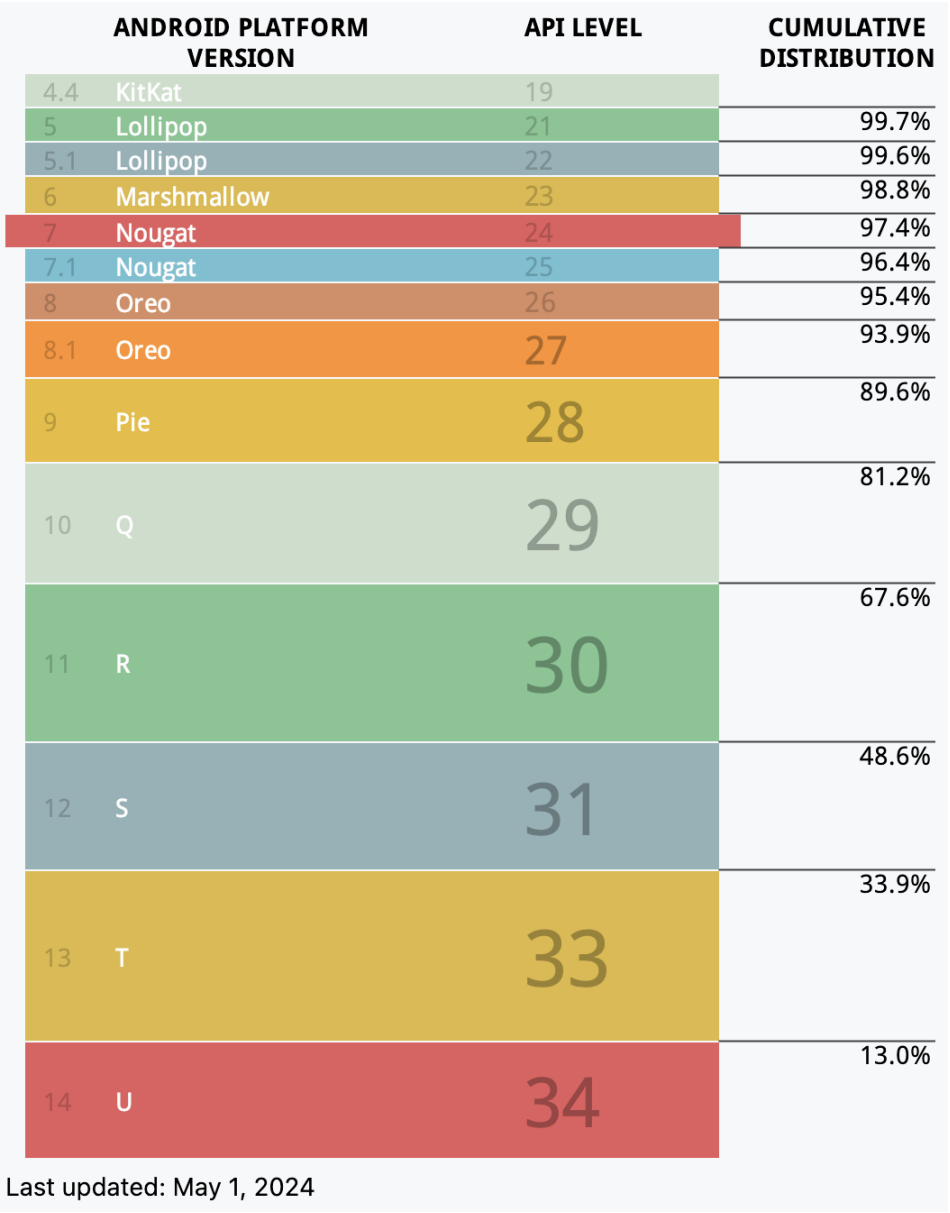


2023.10.04



2024.09.03

Versions



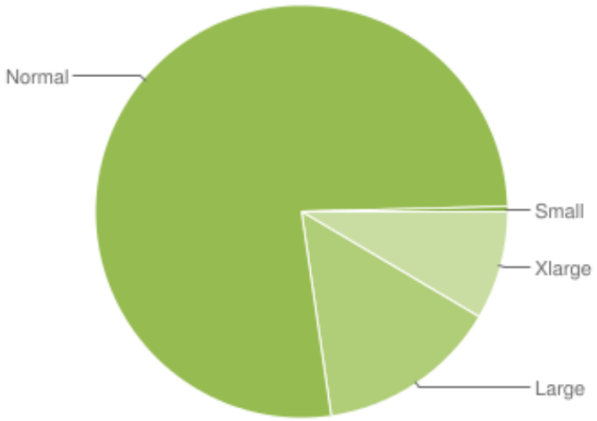
	ANDROID PLATFORM VERSION	API LEVEL	CUMULATIVE DISTRIBUTION
4.4	KitKat	19	
5	Lollipop	21	99.7%
5.1	Lollipop	22	99.6%
6	Marshmallow	23	98.8%
7	Nougat	24	97.4%
7.1	Nougat	25	96.4%
8	Oreo	26	95.4%
8.1	Oreo	27	93.9%
9	Pie	28	89.6%
10	Q	29	81.2%
11	R	30	67.6%
12	S	31	48.6%
13	T	33	33.9%
14	U	34	13.0%

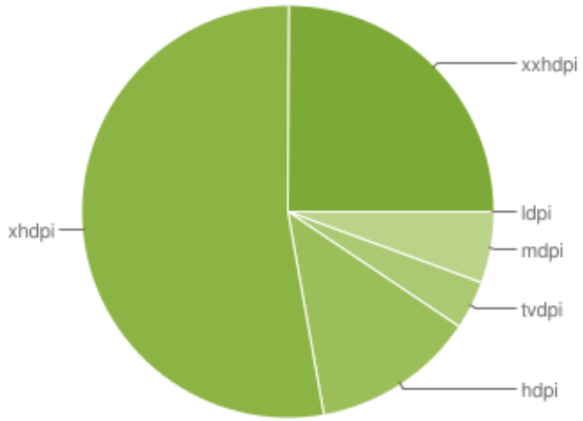
Last updated: May 1, 2024

- Android versions for different devices:
  - Android OS
    - Smartphones, tablets and similar devices
  - WearOS
    - Smartwatches and similar
  - Android TV
    - TV and large screens
  - Android Automotive/Auto
    - AndroidCar
  - ~~Android Things~~
    - ~~IoT~~

Screen resolution

	ldpi	mdpi	tvdpi	hdpi	xhdpi	xxhdpi	Total
Small					0.4%		0.4%
Normal		0.1%	0.3%	7.9%	45.4%	23.2%	76.9%
Large		1.2%	3.4%	1.1%	6.8%	1.7%	14.2%
Xlarge		4.3%	0.1%	3.8%	0.3%		8.5%
Total	0.0%	5.6%	3.8%	12.8%	52.9%	24.9%	



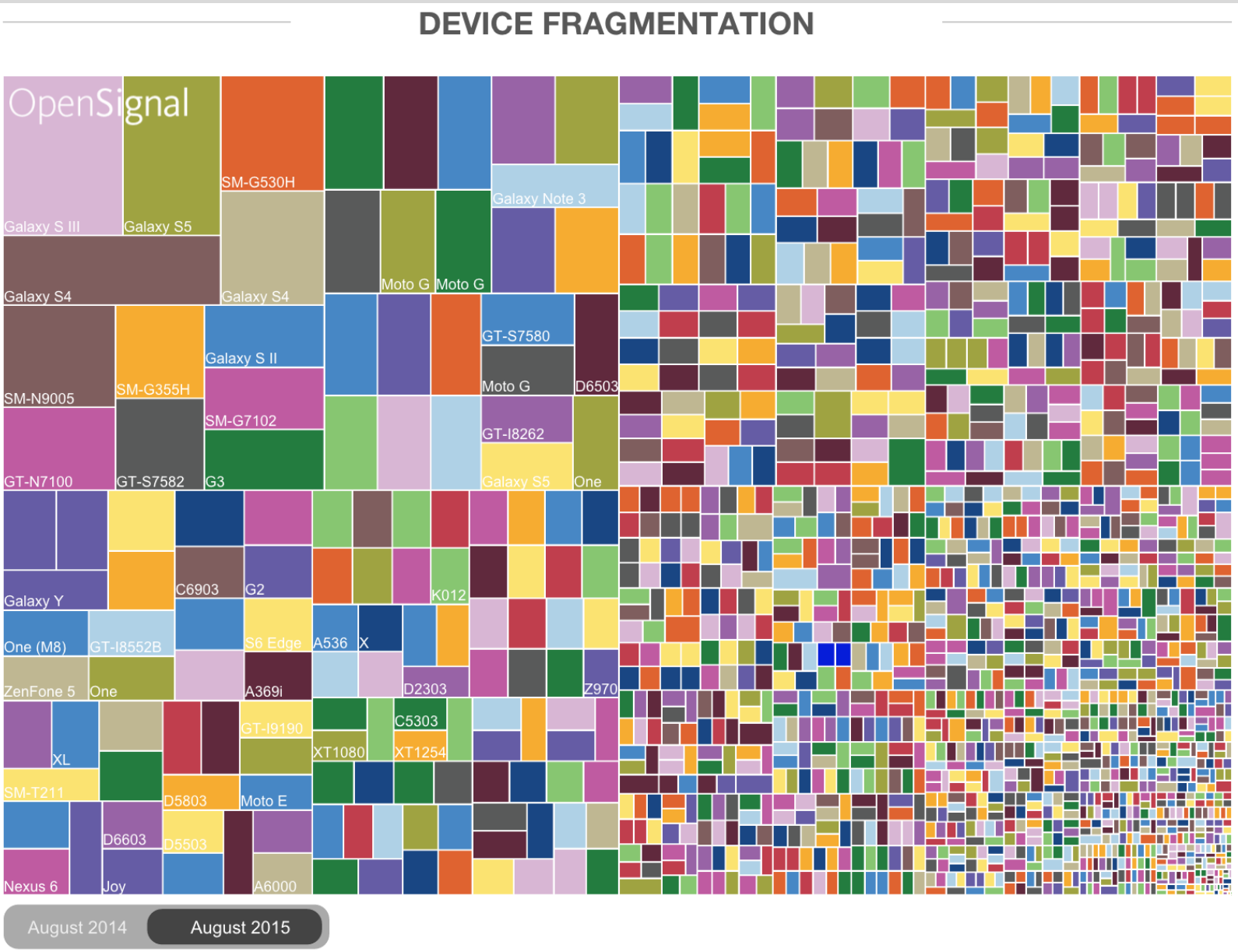


Data collected during a 7-day period ending on January 6, 2023.
Any screen configurations with less than 0.1% distribution are not shown.

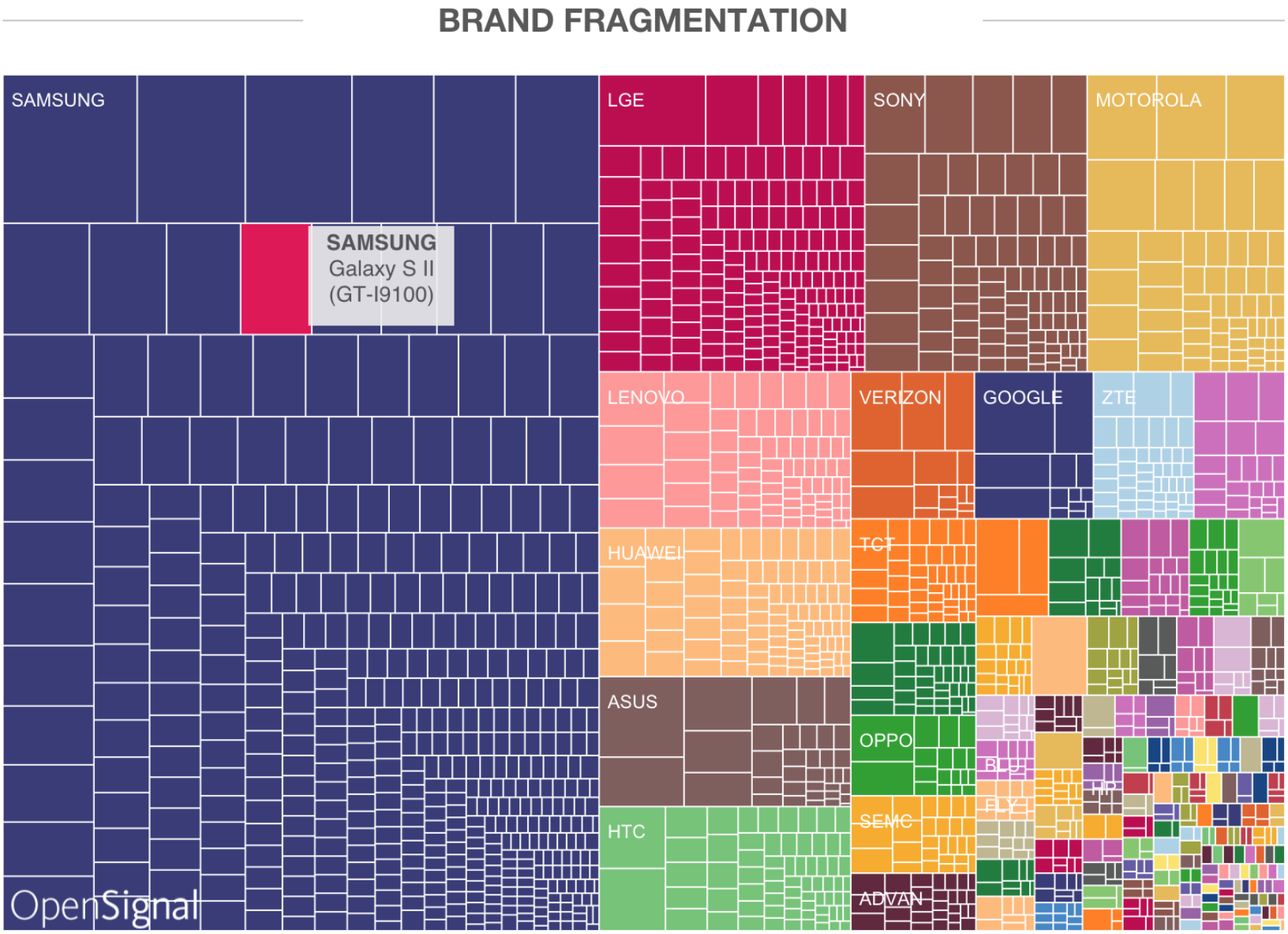
Android Fragmentation

- The problem of fragmentation, usually associated with the Android system, which makes it difficult to create new applications compatible with the various existing devices, arises due to several reasons:
  - Different versions of Android active at the same time
    - Different device manufacturers, with different operating system update policies
    - Economic context of the Android customer opting for cheaper devices
  - Different hardware
    - Screen sizes and resolutions
    - Processors
    - Memory capacity
    - Sensors
    - Communication technologies

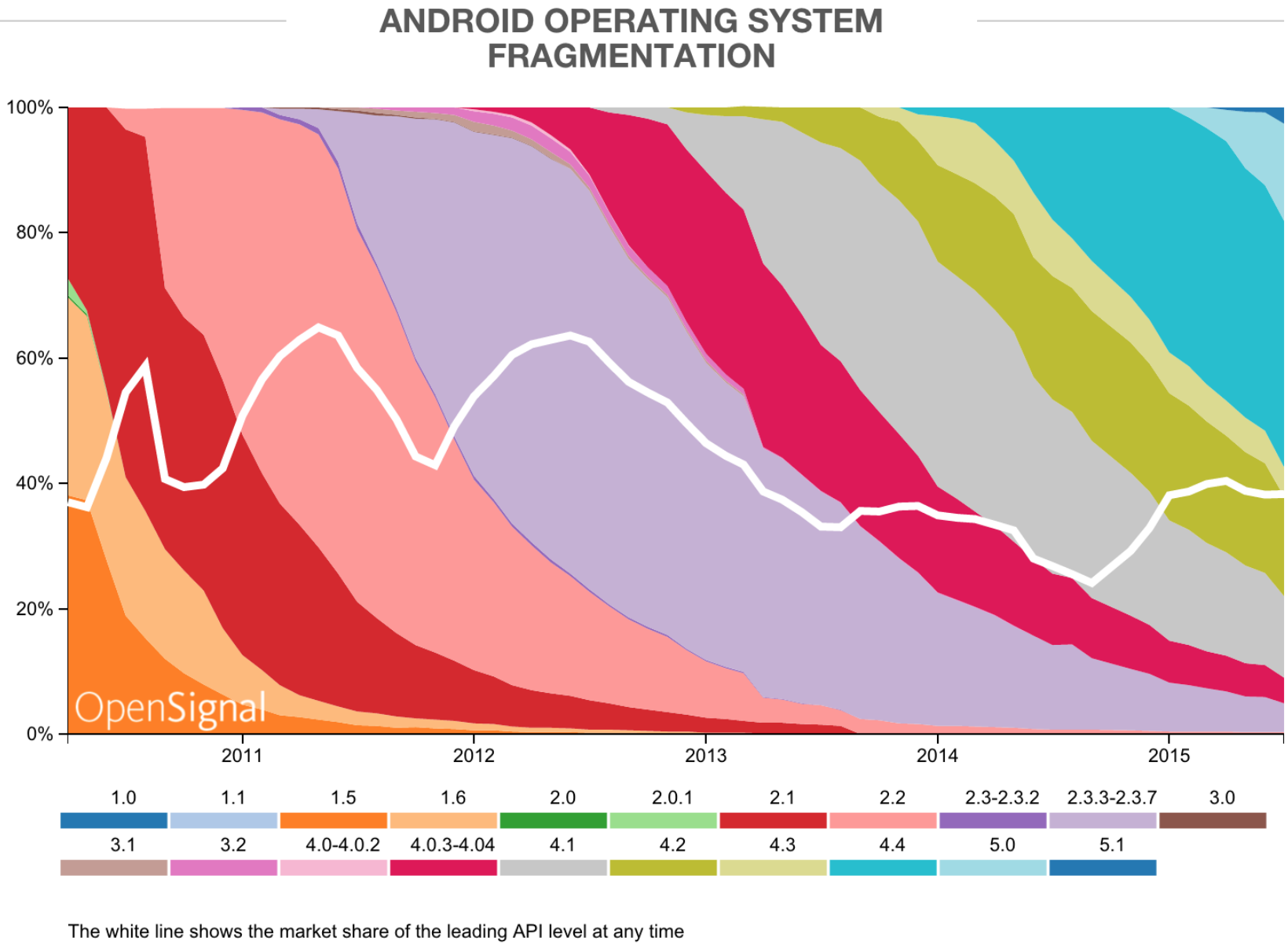
Android Fragmentation



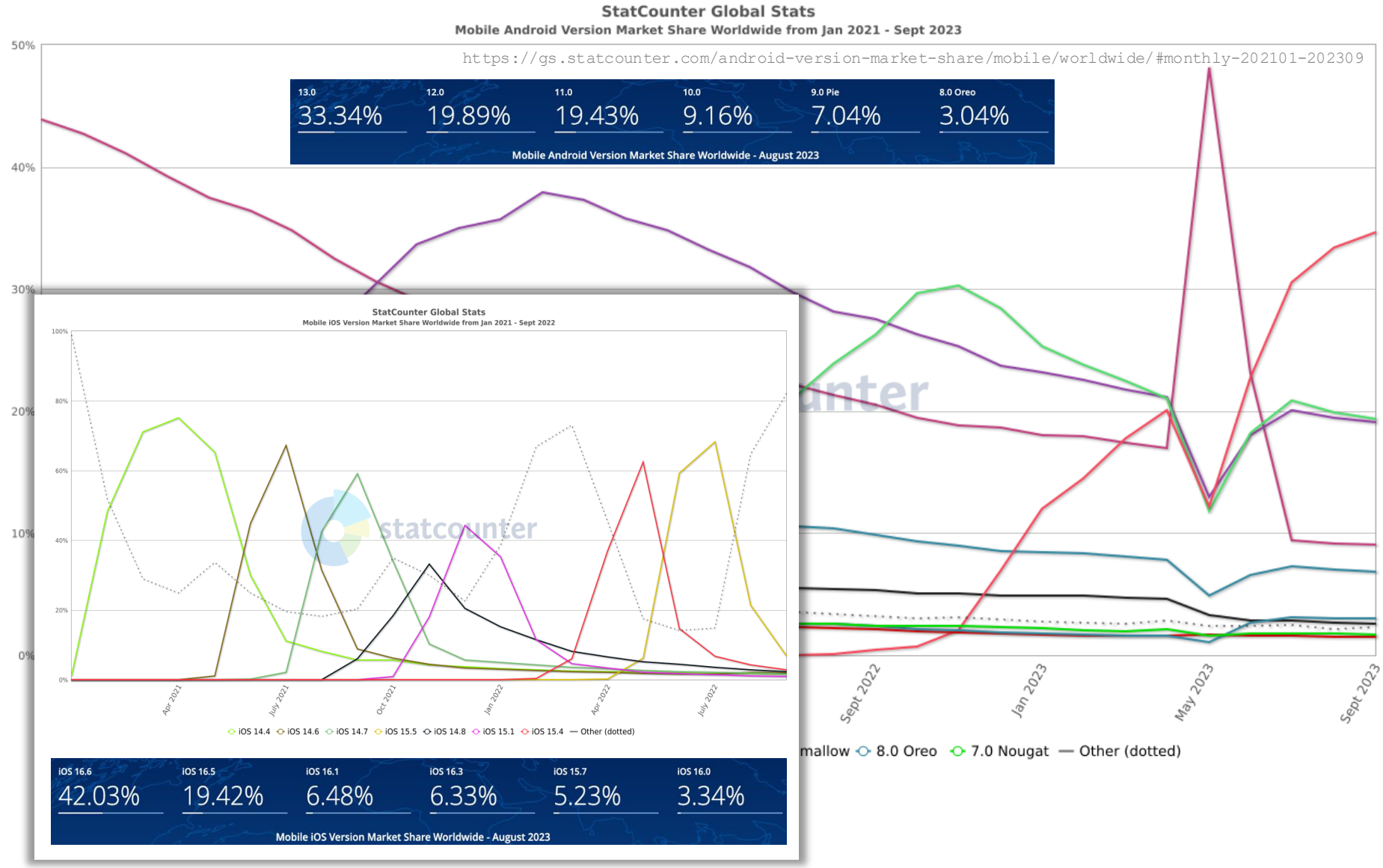
Android Fragmentation



Android Fragmentation



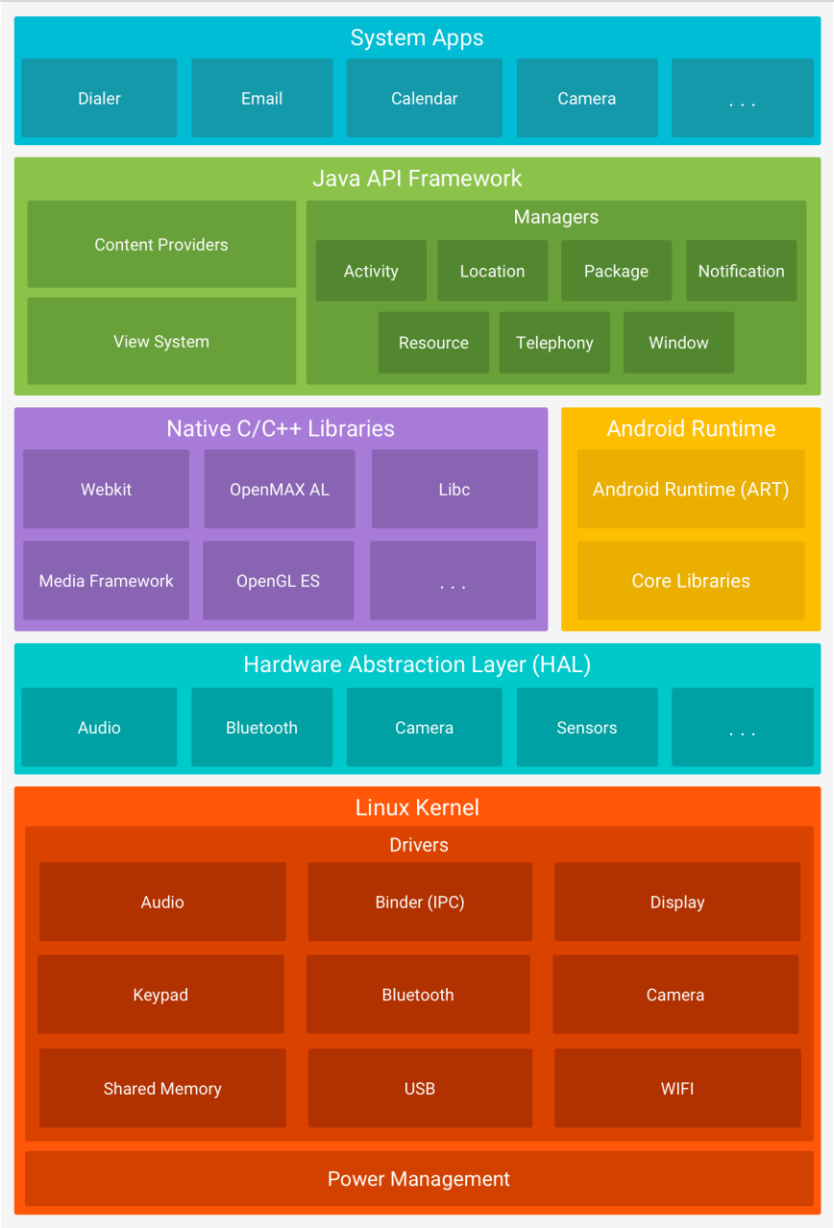
Android Fragmentation



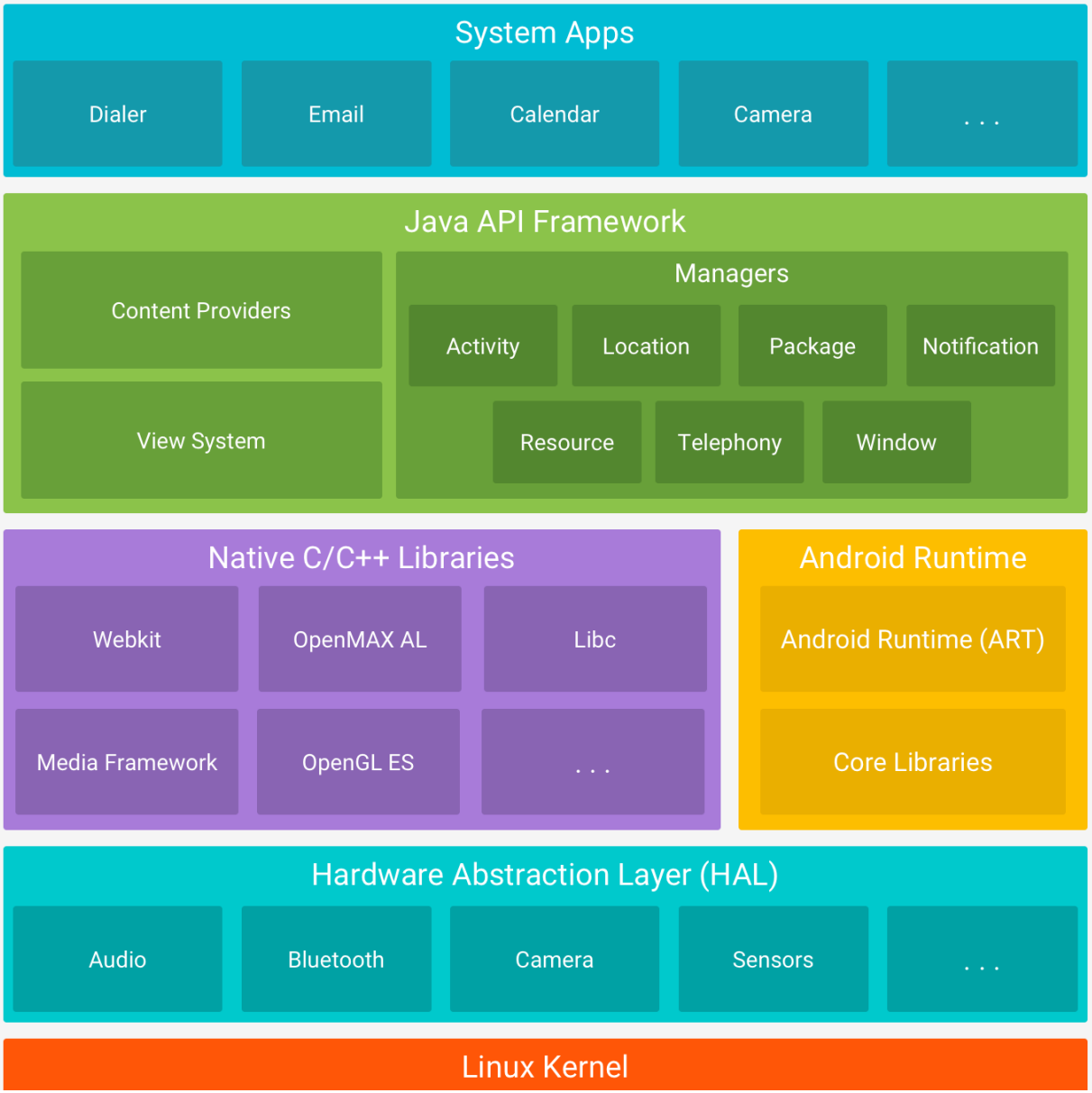
Fragmentation

- Considerations for application development:
  - Different screen dimensions
    - Opt for creating dynamic screens rather than using fixed dimensions
  - Different memory capacities
    - Consider using lower resolution images to support older devices or providing multiple versions of images (with varying resolutions) within the application
  - Different processing capacities
    - Avoid heavy processing within the application; redirect such tasks to online processing whenever possible
  - Dependence on sensors and communication technologies
    - Accelerometers, Gyroscopes, etc.
    - Mobile data, Wi-Fi, Bluetooth, Bluetooth BLE, NFC, etc.
  - Diversity of target audience
    - Account for language and interfaces compatible with RTL (Right-To-Left) text, among other considerations

Android architecture



Android architecture



Android development

- Java, **Kotlin** and C/C++
  - A SDK is provided (Android SDK)
  - IDE: Android Studio, based on IntelliJ by JetBrains
  - Includes emulators for the various versions of Android

- Application distribution
  - packages ".apk"
    - For more recent versions it is mandatory to use bundles ".aab"
  - Google Play Store and other “parallel” markets
  - Packages copied directly to the cell phone can be installed

  - Google Play Store registration has a $25 one-time fee
  - Fee of 30% of the price of each application sold

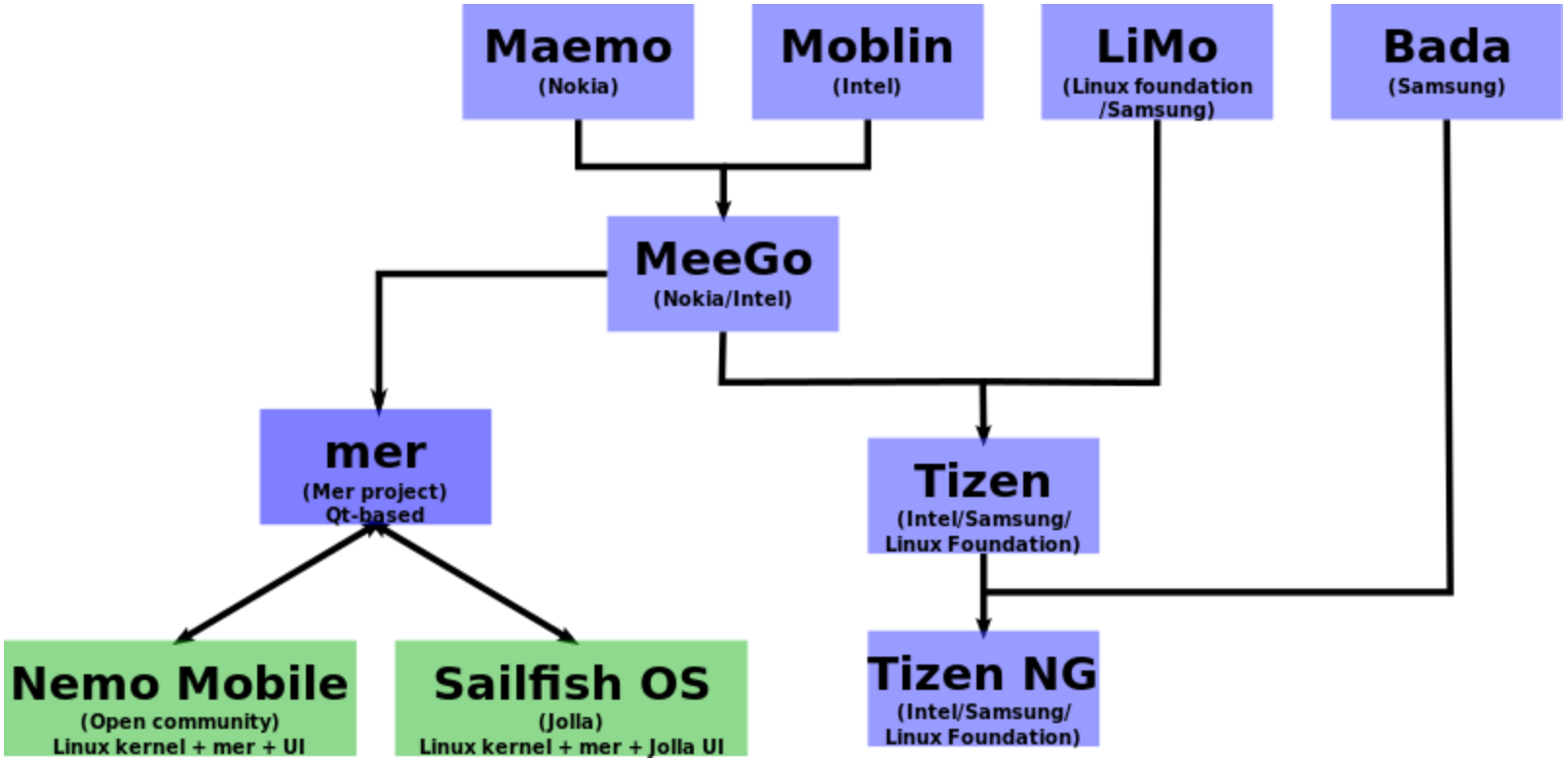
LiMo

- LiMo – *Linux Mobile*
- Management and development by the *LiMo Foundation*
- Essentially focused on *middleware*
- The platform is not *open source* but it is all based on *open source modules*
- Programming/development technologies
  - Java
  - HTML + Javascript
  - C/C++
  - Flash



Tizen , Sailfish ,…

- LiMo + Meego + … => Tizen



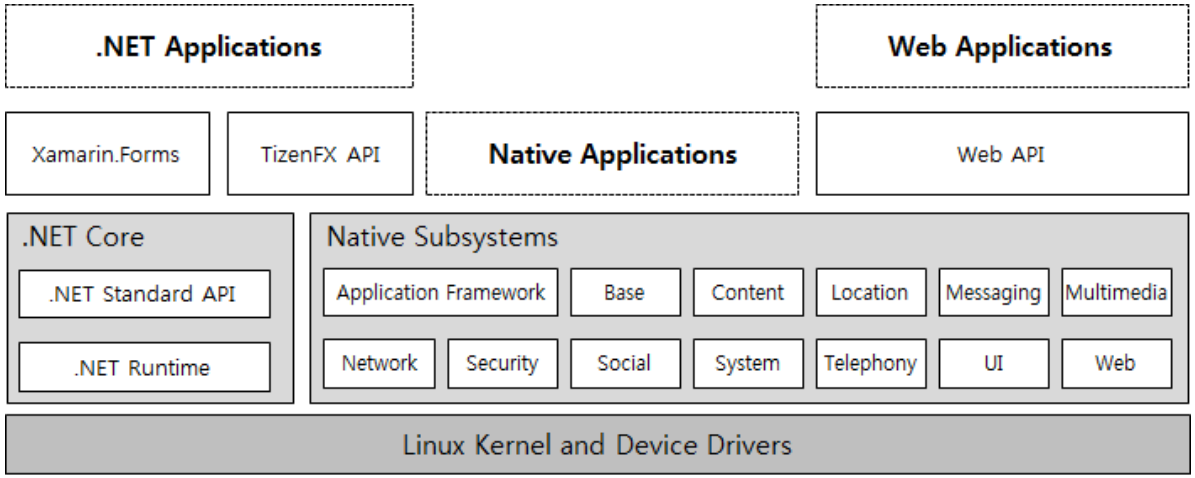
Sailfish OS

- Jolla devices
- Development carried out in:
  - QML (Qt Framework)
  - C++



Tizen

- Project supported by the Linux Foundation
- Development based on
  - Linux kernel
  - GNU C libraries
- Applications developed using
  - Web (HTML5, CSS, JavaScript)
  - Native
  - .NET Applications (C#)

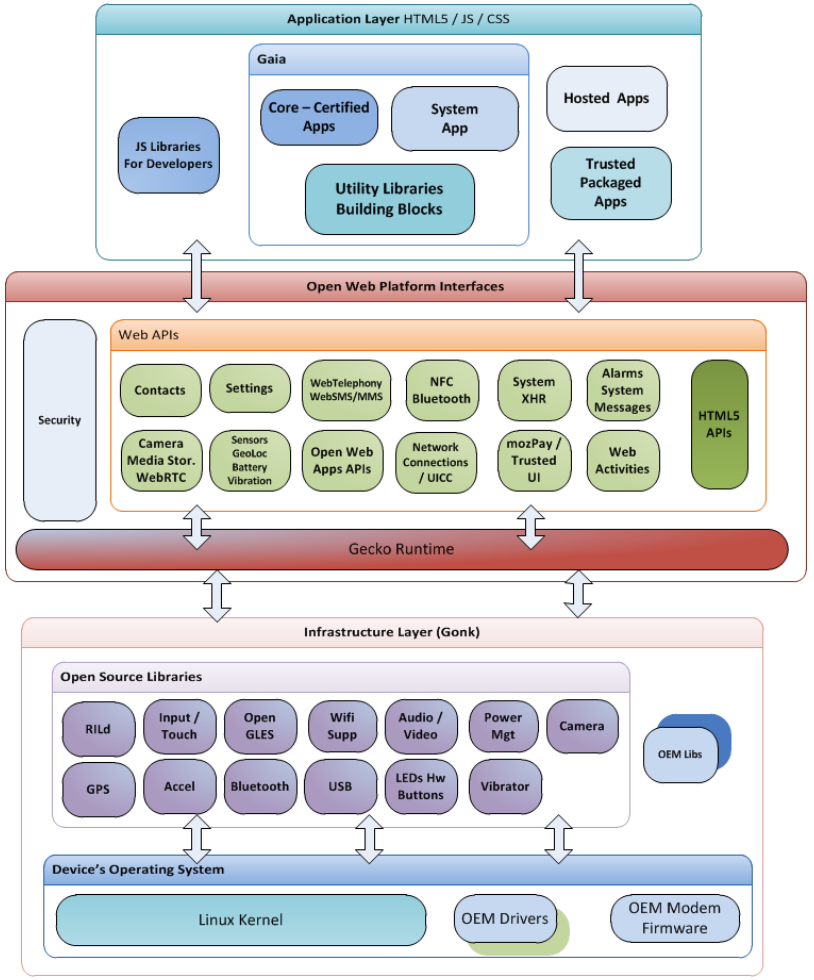


FirefoxOS

- Mainly web-oriented system

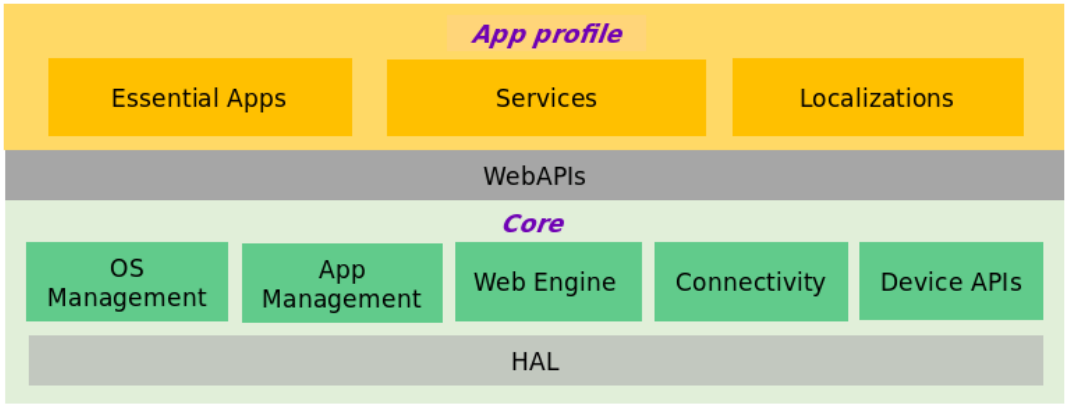
- Use of existing Web standards to provide applications
  - HTML5
  - JavaScript
- Use WebAPIs to access resources available on the hardware

- 1st version: August 2012
- Latest version: May 2016 (version 2.6)



KaiOS

- Linux-based
- Designed for lower-cost cell phones, normally with associated keyboards
- 1st version: March 2017
- Latest version: 3.1 (March 2022)





Cross-platform applications

- Main goal
  - Develop only one version of the application that will run on different operating systems
  - Faster development



Cross-platform

- Benefits
  - Same development language for all systems
  - It is not necessary to have specialized development teams for each system
  - Teams focused on providing new features rather than adapting to different systems
  - Application support made easy
  - Larger target audience

Cross-platform

- Disadvantages
  - Possibly lower application performance
  - Limited access (possibly) to internal equipment resources and sensors
  - Difficulty adapting interfaces to a greater diversity of screen sizes and interaction systems
    - Related to the fragmentation problem mentioned earlier, although existing platforms try to minimize them

Cross-platform

- Until recently, the main solution was the Java language
- There are currently several cross-platform solutions on the market
  - Cross-platform development environments
    - With native code generation for different platforms
    - With object code generation that runs in the context of a pre-installed framework on devices
    - Based on the capabilities of Javascript and HTML5
      - Web Applications
      - Hybrids (native + web)



